

HW2 - Design Areas

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Collaborators: N/A

Idea

Personalized AI Study Companion - a mobile and web app powered by generative AI. It helps students learn more effectively through creating dynamic study plans, adaptive quizzes, AI tutoring sessions, and real-time progress feedback. Unlike static flashcards or generic video platforms, this tool tailors itself to each student's learning style and goals.

Similar Existing Services

- **Quizlet** - flashcards, but largely user-generated and non-adaptive.
- **Khan Academy** - excellent structured lessons but not personalized per learner.
- **ChatGPT** - can answer questions, but lacks progress tracking and targeted study flows.

Differentiation: This service will merge real-time generative AI with adaptive learning. It continuously evaluates what the student knows, generates tailored content, and updates study strategies. This only recently became feasible because of advances in generative AI models that can create high-quality, context-specific learning content at scale.

Stakeholders

- **Students (primary users):** Want effective, personalized study help. Motivated by academic performance and time efficiency.
- **Teachers/Educators:** Want a supplemental teaching tool but may want oversight of generated content. Motivated by student success.
- **Parents:** Want reliable tools that improve their children's learning outcomes. Concerned with cost, privacy, and safety.
- **Investors:** See opportunity in edtech market growth. Motivated by revenue, uniqueness, and scalability.
- **Regulators:** Care about data privacy (especially that of minors), algorithmic transparency, and ethical use of AI.
- **Developers/Engineers:** Focus on creating robust, accurate, and user-friendly systems.

User Persona & Journey

Persona: Jack, the Student

- **Backstory:** A 19-year-old college freshman majoring in biology. Struggles with keeping up in organic chemistry. Balances classes, part-time work, and extracurriculars.
- **Values:** Efficiency, clarity, and confidence in mastering material.
- **Motivations:** Wants to improve grades without burning out.
- **Goal:** “I want a study system that adapts to me, so I don’t waste time studying the wrong way.”

Current Journey (Without the AI Companion)

1. Jack logs into multiple resources (Canvas, Quizlet, YouTube) to find study material.
2. He spends 20 minutes searching for good explanations of a confusing concept.
3. He manually makes flashcards but struggles to know which areas to focus on.
4. After studying for hours, he still feels unsure about what he actually knows.
5. Before the test, he crams, repeats content he already mastered, and misses weak spots.

Pain Points:

- Too much time wasted finding and curating resources.
- Anxiety about not knowing if he’s truly prepared.
- Generic content that doesn’t fit his learning gaps.

Improved Journey (With the AI Companion)

1. Jack logs into the AI Study Companion with his student account.
2. The system authenticates him and asks about his courses and upcoming exams.
3. He selects the folder for the course “Organic Chemistry”, uploads the relevant notes or materials (if any), and specifies: “Exam in 10 days, I struggle with resonance structures.”
4. The AI generates a personalized study plan with daily review chunks and quizzes.
5. Jack starts a study session. The AI tracks his responses, identifies that he’s strong on mechanisms but weak on electron movement, and dynamically adjusts the material.
6. The AI tutor simulates an exam-style Q&A session, highlighting the specific regions where he struggles.
7. The app provides spaced repetition reminders and confidence metrics (“You’ve improved 40% in resonance problems since yesterday”).
8. Jack feels confident and balanced, and he no longer has to cram.

The improvements include

- **Efficiency:** Jack no longer wastes time searching for study material; the app curates it instantly.
- **Personalization:** The AI detects his weak areas and tailors quizzes/explanations to those gaps.
- **Clarity:** Real-time feedback shows his progress, reducing test anxiety.
- **Retention:** Adaptive spaced repetition reinforces long-term memory.
- **Motivation:** Progress tracking and confidence boosts keep him engaged.

Values Tensions in Jack's Journey

1. Jack (Student) vs. Teachers/Educators

- Jack's value: Desires immediate, adaptive help based on his weaknesses, even if the explanations are crude.
- Teacher's value: Desires challenge and alignment with course objectives, afraid the AI will "oversimplify" or misrepresent challenging material or that it might do the student's homework for them.
- **Tension:** The AI must balance personalization and scholarly integrity to the teacher's curriculum.

2. Jack vs. Parents

- Jack's value: Desires autonomy and privacy; may not wish parents to be aware of all his challenges.
- Parent's value: Wants insight into child's learning path and make sure the app is valuable and safe enough to pay for.
- **Tension:** Parents can push for monitoring dashboards, and Jack can push back on monitoring to stay independent.

3. Jack vs. Regulators

- Jack's value: Wants the most extreme personalization, even if that means the system is collecting a lot of personal data.
- Regulator's value: Wants minimization of data and compliance (FERPA, GDPR, COPPA if under 18).
- **Tension:** Regulators may restrict the amount of adaptive tracking allowed, perhaps limiting the AI from personalizing study routes.

4. Jack vs. Investors

- Jack's value: Wants affordability and openness (ideally free or low-cost).
- Investor's value: Profit-maximizing, perhaps through subscription, advertising, or paywalls.
- **Tension:** Push to monetize might make the tool less available to under-resourced students.

5. Jack vs. Developers/Engineers

- Jack's value: Wants effortless, effortless, and immediate functionality.
- Developer's value: Must balance innovation against technical constraints (model cost, computing power, glitches).
- **Tension:** Possibilities Jack wants (e.g., real-time AI tutoring) may prove difficult to deliver consistently, and this creates frustration.